

GmailGuard

Working Plan & Technical Architecture

Smart India Hackathon 2026

SIH26106 — AI-Powered Email Threat Detection, GeoLocation, and Forensic Intelligence Platform

Core Idea

GmailGuard analyzes an email using multiple evidence sources — authentication, observable infrastructure, IP/domain intelligence, URLs, attachments and NLP/ML — and correlates them into an explainable threat assessment.

Design principle: No single signal determines whether an email is safe. The system correlates independent evidence and explains the result.

1. End-to-End Working

Stage	Working	Output
Email input	Read raw .eml/email data	Structured email
Parsing	Extract headers, body, HTML, attachments	Parsed components
Header & auth	Analyze Received, SPF, DKIM, DMARC	Authentication + infrastructure evidence
Infrastructure	Extract public IPs; identify network/provider; geolocate when possible	Forensic context
Content	Run TF-IDF + Linear SVM	Phishing / Legitimate
URLs & files	Static analysis only	Risk indicators
Correlation	Combine available signals	Threat score
Reporting	Explain findings and limitations	Analyst-friendly report

2. ML Work Completed

Dataset: phishing_email.csv from the selected Kaggle phishing email dataset. Missing/empty records and duplicates were removed; data was split using a stratified 80/20 train-test split.

Feature extraction: TF-IDF with up to 50,000 features. The vectorizer was fitted only on training data and then used to transform test data.

Models compared: Logistic Regression, Multinomial Naive Bayes, and Linear SVM.

Model	Accuracy	Decision
Logistic Regression	≈98.59%	Compared
Naive Bayes	≈97.67%	Compared
Linear SVM	≈99.21%	Selected

Tuning: GridSearchCV tested C = [0.1, 0.5, 1, 2, 5]. Best C = 1. Test accuracy remained approximately 99.21%.

Evaluation: accuracy, precision, recall, F1-score and a confusion matrix were reviewed. The selected test split had 85 false positives and 45 false negatives.

Explainability: SVM coefficients were inspected for phishing- and legitimate-associated text features. These are model associations, not causal proof.

Saved artifacts: gmailguard_svm.pkl and gmailguard_tfidf.pkl. Both are required for inference.

3. Current ML Architecture

Raw email text → TF-IDF → Linear SVM (C=1) → Phishing / Legitimate.

The SVM is the content-analysis component, not the entire GmailGuard decision system.

Important: LinearSVC decision scores must not be presented as calibrated probabilities unless calibration is added later.

4. Python Modules and Responsibilities

File	Working
email_parser.py	Reads .eml data and extracts sender, recipient, subject, headers, body, HTML and attachments.
header_analyzer.py	Analyzes technical headers such as Received, Return-Path and Authentication-Results; identifies observable IP candidates
authentication_analyzer.py	Handles SPF, DKIM and DMARC results as PASS/FAIL/NONE/UNKNOWN.
ip_intelligence.py	Classifies IPs and supports network/ASN/provider and reputation intelligence.

File	Working
geolocation.py	Maps observable public IPs to geographic/network context; does not claim exact attacker location.
domain_intelligence.py	Analyzes sender/extracted domains and provides a place for reputation intelligence.
url_analyzer.py	Extracts URLs and performs safe static checks for suspicious characteristics.
attachment_analyzer.py	Safely inspects filename, extension, MIME type, size and suspicious static indicators; never executes files.
ml_classifier.py	Loads the saved TF-IDF vectorizer and Linear SVM for new-email prediction.
threat_scorer.py	Correlates ML, authentication, URL, attachment, IP and domain evidence into a configurable risk score.
report_generator.py	Creates a human-readable investigation report with verdict, evidence and limitations.
main.py	Orchestrates the pipeline and can later serve as an application/API entry point.
config.py	Centralizes paths, scoring weights, provider settings and configuration.
requirements.txt	Lists Python dependencies.

5. IP Intelligence & Geolocation

Purpose: provide forensic context about observable email infrastructure, not to claim the sender's physical location.

Workflow: **Received headers** → **public IP extraction** → **IP classification** → **network/provider intelligence** → **GeoIP lookup** → **infrastructure location**.

- A visible IP may belong to AWS SES, Google, Microsoft, a company mail server, or another relay.
- No observable sender/device IP is also a valid result; the system should report the limitation and continue with other evidence.
- Geolocation itself should have limited direct influence on the threat score.

Judge-ready statement: "We use geolocation as a forensic intelligence layer to understand the geographic and network context of observable email infrastructure. We do not claim that the mapped location is the attacker's exact physical location."

6. Threat Scoring & Correlation

The correlation engine is the layer that converts separate findings into one explainable risk assessment. Initial weights are configurable starting points, not claims of scientific optimization.

Signal group	Initial weight
NLP / ML	30%
SPF / DKIM / DMARC	20%
URL analysis	20%
IP / network intelligence	10%
Domain intelligence	10%
Attachment analysis	10%

- UNKNOWN information must not automatically be treated as malicious.
- Authentication PASS provides positive evidence but does not guarantee safety.
- Suspicious/malicious URLs, attachments, domains, authentication failures and bad reputation can increase risk.
- Every final score should include the evidence that contributed to it.
- Geolocation primarily supplies forensic context.

7. Example Final Result

Field	Example
Threat Score	82 / 100
Verdict	HIGH RISK
ML	Phishing
Authentication	SPF PASS / DKIM PASS / DMARC FAIL
URL	Suspicious
Attachment	Suspicious
Infrastructure	Observable IP + provider + GeoIP context
Limitations	Sender device IP not observable

8. Testing Structure

- test_email_parser.py — parser tests
- test_authentication_analyzer.py — SPF/DKIM/DMARC tests
- test_attachment_analyzer.py — safe attachment-analysis tests
- test_url_analyzer.py — URL extraction/analysis tests
- test_ml_classifier.py — TF-IDF + SVM loading/prediction tests
- test_threat_scorer.py — scoring and evidence tests
- test_integration.py — end-to-end module integration tests

9. Development Roadmap

Area	Status
Dataset + ML pipeline	COMPLETED
Model tuning/evaluation/explainability	COMPLETED
Saved model artifacts + prediction function	COMPLETED
Email parser / header analysis	IMPLEMENT / VERIFY
SPF/DKIM/DMARC	IMPLEMENT / VERIFY
IP intelligence + geolocation	IMPLEMENT / VERIFY
URL + attachment analysis	IMPLEMENT / VERIFY
Threat correlation/scoring	IMPLEMENT / VERIFY
FastAPI integration	NEXT
Frontend / analyst dashboard	NEXT
End-to-end demo/testing	NEXT

10. Judge-Friendly Explanation

30-second pitch: “GmailGuard is not just a phishing classifier. We analyze an email from multiple angles: sender authentication, observable mail infrastructure and IP intelligence, domains and URLs, attachments, and AI-based content analysis. We then correlate these signals into an explainable threat score. If an IP is visible, we can provide infrastructure geolocation; if it is not visible, we report that limitation instead of inventing a location.”

Why geolocation? “Geolocation is a forensic intelligence layer. It helps us understand the geographic and network context of observable email infrastructure. It is not used alone to declare an email safe or to claim the attacker's exact physical location.”

Why AWS/SMTP does not mean safe? “SMTP and cloud email services describe how an email was delivered, not whether the sender is trustworthy. That is why we correlate infrastructure evidence with authentication, reputation, URLs, attachments and ML.”

11. Final Architecture

Raw Email → Parser → Header & Authentication Analysis → IP/Domain/Geolocation Intelligence + URL/Attachment Analysis + NLP/ML → Correlation/Threat Scoring → Explainable Report → FastAPI → Frontend.

This document reflects the current GmailGuard working plan discussed so far. Implementation status should be verified against the actual source code before final presentation claims.